The Islamia University of Bahawalpur

Final Year Project Proposal

Program:

[x] BSIT **[ ]** BSSE **[ ]** BSAI

Project Title: TaskKaro AI: An AI-Powered Voice-Enabled Gig Marketplace with Intelligent Task Matching

Nature of Project:

[x] Development **[ ]** Research

Area of Specialization: (Only relevant areas are selected)

[x] Artificial Intelligence (AI) **[ ]** Game Development **[ ]** Internet of Things (IoT) **[ ]** Machine Learning **[ ]** Mobile App Development **[ ]** Blockchain **[ ]** Data Science and Analytics [x] Web Development **[ ]** Cybersecurity **[ ]** Image Processing **[ ]** Desktop Application [x] Natural Language Processing (NLP) **[ ]** Other

Introduction

TaskKaro AI is a web-based gig marketplace designed to connect customers who need everyday errands completed with people seeking flexible earning opportunities. Customers can describe tasks through text or voice in English, Urdu, or Roman Urdu. An AI agent interprets the request, collects missing details, and converts it into a structured task. The platform then helps identify suitable workers based on skills, location, availability, and task requirements. Example tasks include grocery shopping, parcel collection, and luggage pickup or delivery. The system combines AI-assisted request understanding and matching with backend rules for verifying availability, assigning tasks, and maintaining records.

Problem Statement

People often rely on informal contacts or separate service providers to get everyday errands completed. This can make it difficult to describe a task clearly, find an available suitable person, and keep track of progress. At the same time, people seeking small flexible gigs may have difficulty discovering relevant tasks. Common limitations this project aims to address include:

• Requests may be shared through unstructured messages or calls, making task details easy to miss. • Customers may need to contact multiple people to find someone with the right skills and availability. • Language barriers and typing difficulty can make digital task booking less accessible.

• Customers and workers may lack one place to view task status, assignment details, and earnings records.

TaskKaro AI will address these issues through voice/text request understanding, follow-up questions, structured task creation, AI-assisted worker matching, and centralized task tracking. The project is not intended to guarantee worker reliability or automate real-world safety checks; the platform will provide a basic workflow that can be tested with sample users and data.

Project Goal / Objectives

The main goal is to develop an AI-powered gig marketplace that simplifies everyday task booking and helps connect customers with suitable workers.

1. Develop customer, worker, and admin interfaces for managing accounts and tasks.

2. Enable customers to submit task requests using text or voice in English, Urdu, or Roman Urdu, subject to the selected speech services.

3. Use an LLM API to extract task details into a structured format and ask for missing information. 4.

Rank potentially suitable workers using task requirements, skills, location, and availability. 5. Use

backend validation to confirm worker availability and control task assignment and status updates. 6.

Allow workers to view and accept offered gigs, update task progress, and view earnings records. 7.

Provide an admin dashboard to manage users, workers, task records, and platform activity. 8. Test the

main workflows using sample tasks and document the results and limitations.

Application Architecture / Technology Stack / Tools / Hardware

Architecture: Client-server web application with a frontend, FastAPI backend, database, and external AI/speech service APIs.

Frontend: React.js or Next.js with JavaScript/TypeScript, HTML, and CSS.

Backend: Python with FastAPI; REST API endpoints and server-side validation.

Database: SQLite for the initial prototype or PostgreSQL for a more production-oriented setup.

AI / NLP: LLM API for natural-language understanding, structured task extraction, follow-up question support, and matching assistance.

Voice: Speech-to-text and text-to-speech services selected during implementation.

Tools: Visual Studio Code, Git, GitHub, API testing tools, and n8n (optional for workflow automation).

Hardware: A development laptop/PC with internet access; server/cloud hosting may be used for deployment. Exact requirements depend on the chosen AI and speech APIs.

Important system boundary: AI will interpret requests and assist with worker ranking. The backend—not the AI model—will validate availability and enforce assignment, task-status, and earnings-record rules.

Project Member

Sr. #	Registration #	Student Name	Phone #	Section
1	F23BINFT1M01066	ZUHA NOOR	03109522406	7TH sem 2M

References (if any)

No external system is claimed as the direct basis of this proposal. The problem description is based on the intended use case of informal everyday-task coordination and the project's planned features. Add verified references if your supervisor requires a competitor or research comparison.

Project Supervisor

Name: Dr. Rana Nazeer Ahmad

Supervisor Acceptance Signature: _____

Head of Department (HoD): _____